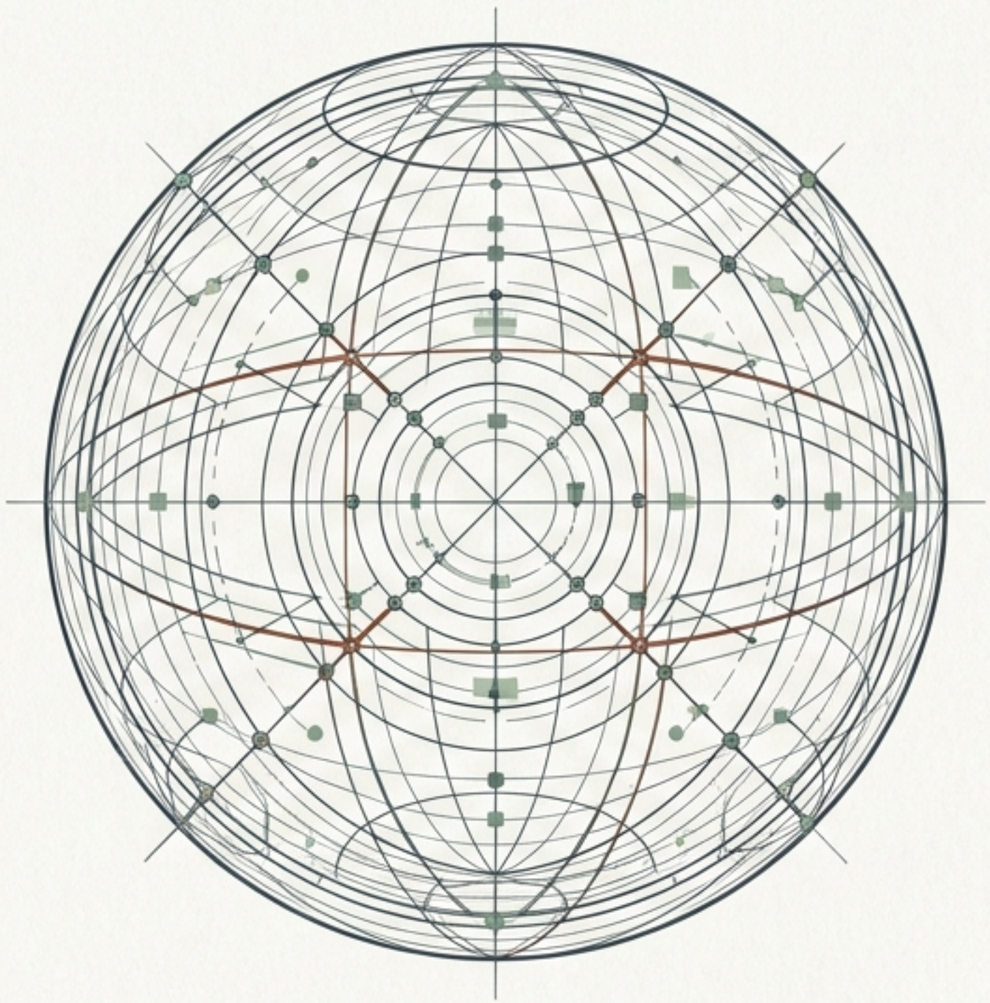
BUYWISE AI

Know Before You Buy.

An AI-powered smart product decision assistant architected specifically for the nuances of the Indian Consumer Market.



Gemini Notebook

Navigating the Indian E-Commerce Minefield



The Intelligence Layer

Intercept

AI-driven product searches replace manual scrolling.

Evaluate

Calculates honest, algorithm-backed Trust Scores.

Protect

Flags synthetic reviews and artificially inflated ratings.

Contextualize

Checks current market pricing (₹) against active alternatives.

Gemini Notebook

The Intelligent Consumer Frontend



Real-Time Intelligence

AI-powered product analysis complete with a dynamic trust score ring and real-time live web fallbacks.



Objective Comparisons

Side-by-side automatic extraction of specs, TF-IDF generated Pros/Cons, and an LLM-synthesized verdict naming the definitive Better Choice.

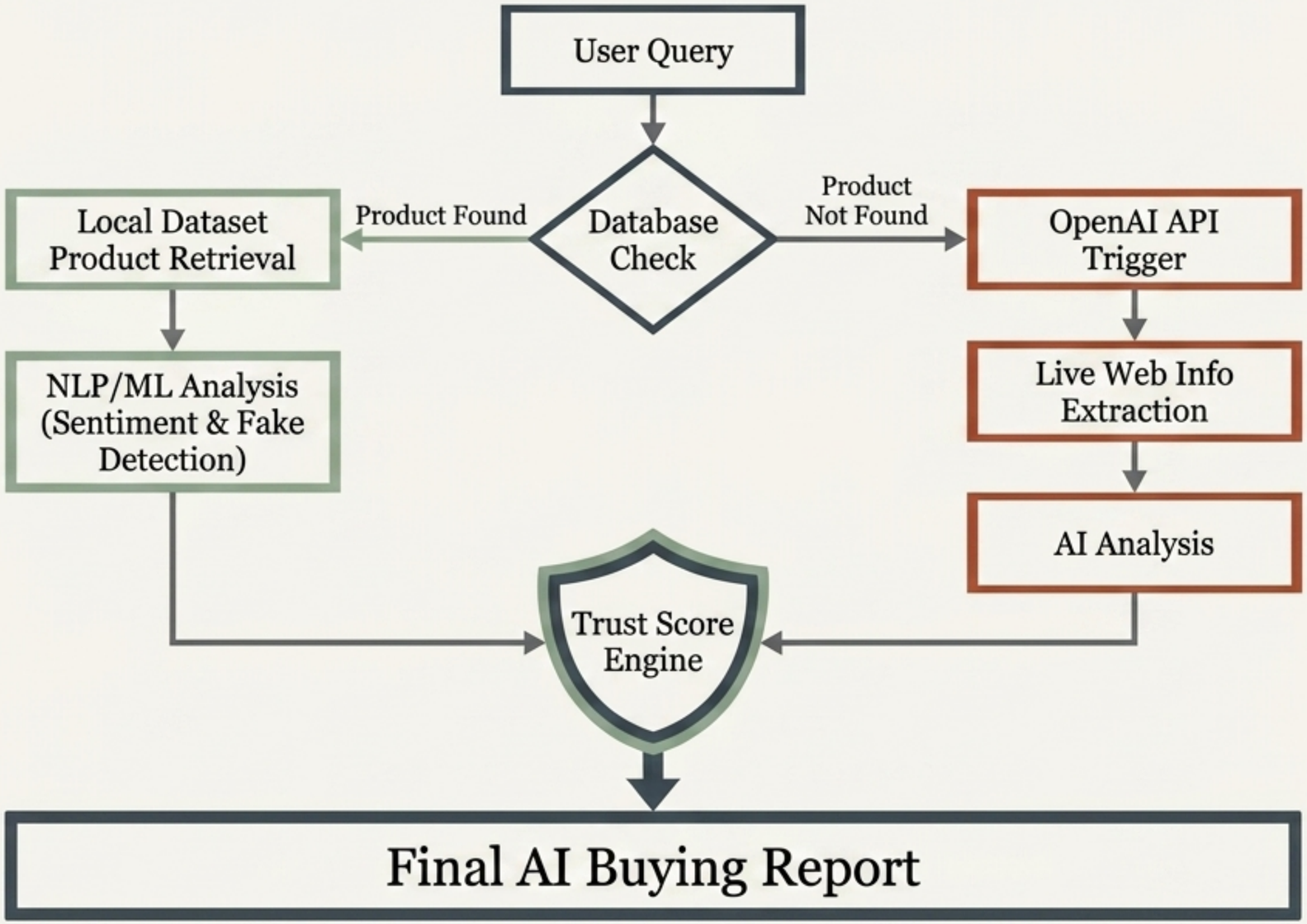


Personalized Consultation

A RAG-powered AI assistant that dynamically generates tailored questions grounded in product context to calculate an exact user Match Percentage.

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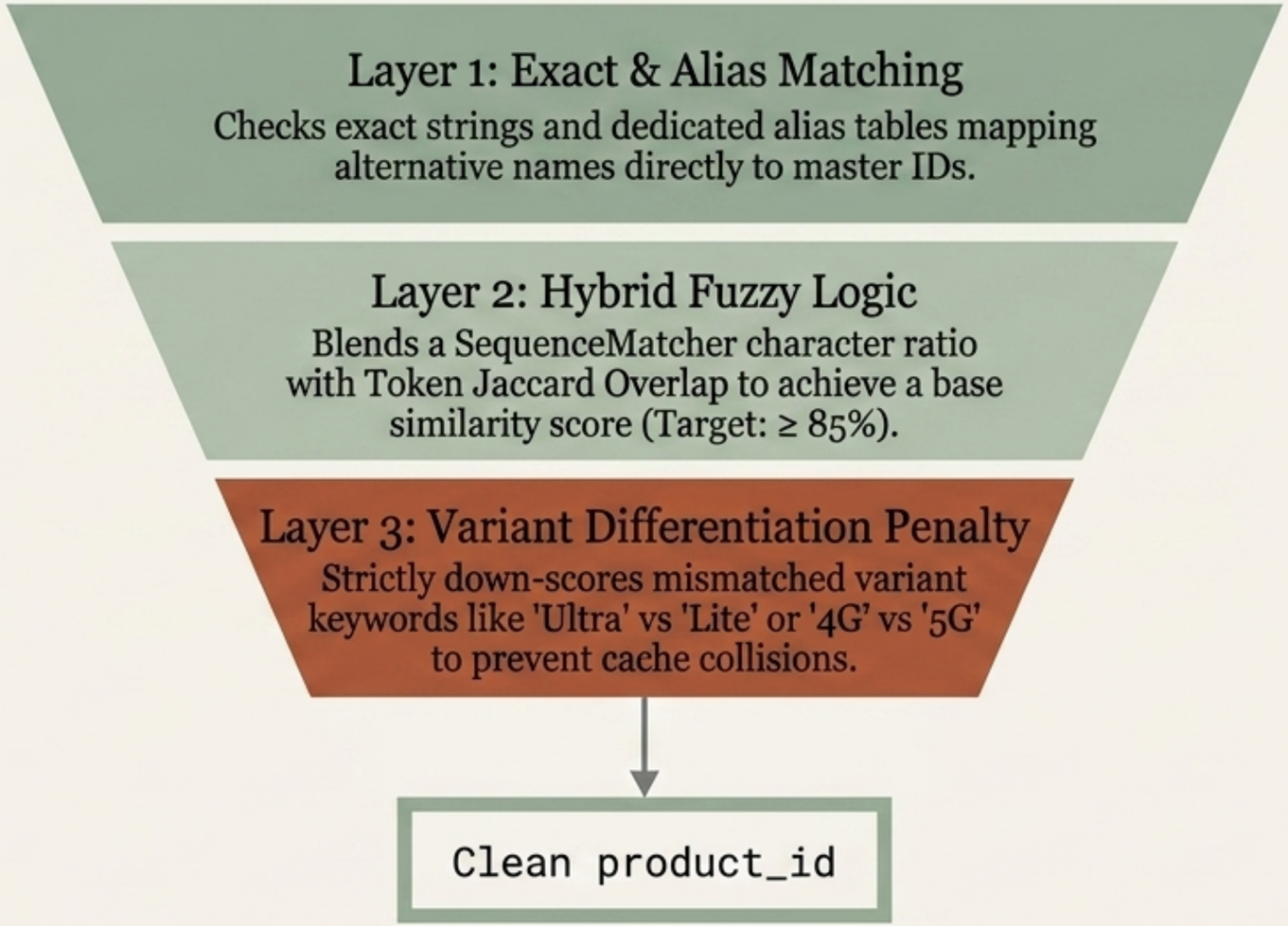
Dual-Path Processing Architecture



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Search Resolution and Fuzzy Matching

Galaxy S23 Ultra samsung s23 lite ASIN: B0BY8L



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The Trust Score Equation

$$(0.40 \times \text{Base Sentiment}) + (0.30 \times \text{Normalized Rating}) + (0.20 \times \text{Spam Cleanliness}) + (0.10 \times \text{Verified Bonus})$$

Base Sentiment	Normalized Rating	Spam Cleanliness	Verified Bonus
NLTK VADER NLP compound scoring across the review pool.	Standard 5-star rating converted to a 100-point scale.	Deductions driven by the fake review detection engine.	Boost for authenticated community or dataset reviews.

THE OBSOLESCENCE OVERRIDE

If the system detects discontinued status, obsolete hardware (e.g., 28nm chips), or dead software (Android 7), the Trust Score is aggressively capped at a maximum of 45/100, overriding historical 5-star reviews.

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Diagnostic Fake Review Detection

Symptom	Detection Method	Suspicion Penalty
Exact Duplicate	MD5 hashing of normalized text	+0.50
Near-Duplicate	TF-IDF Vectorizer + Cosine Similarity (≥ 85%)	+0.30
Sentiment Mismatch	Flags 5-star ratings with VADER compound ≤ -0.3, or 1-star ratings with VADER ≥ 0.3	+0.40
Low-Information / Spam	Regex checks for excessive punctuation (!/?) or reviews under 5 words	+0.20

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Synthesizing the AI Buying Report

Raw Intelligence

- Raw Review JSON Array
- TF-IDF Extracted Pros/Cons n-grams
- VADER Sentiment Percentages
- Live Web Pricing Data

The Persona Engine: GPT-4o prompted strictly as an Indian Consumer Advocate.

Structured Output

3-Sentence Executive Summary

- Sentences are bravnsincensive sentences.
- Sentences about our strategic-trimary product.
- Sentences are unimaver product drattments.

Final Strategic Verdict

Final Strategic Verdict is confised with matlcn an Indian Consumer Advocate.

Current Market Price Justification

Reference, Current Market Price ir explananting the market data, and value proposition from our value proposition to references.

2-3 Highly Specific Alternatives in the Exact Price Band

→ Product Name 1 - Description
→ Product Name 2 - Description
→ Product Name 3 - Description

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Two-Stage Personalized Consultation



Stage 1: Dynamic Intake (gpt-4o)

Generates 10 tailored, category-specific questions.

Example: Asking about display refresh rate preferences for phones, or ANC needs for audio, rather than generic queries.



Stage 2: Recommendation Synthesis

Evaluates the user's 10 specific answers against the product's hardware specs and live market alternatives.

The Verdict

Outputs an exact match percentage, lists why it fits, identifies specific user drawbacks, and curates tailored alternatives.

Gemini Notebook

The Hybrid Data Engine

Admin Toggle Control: Dynamically route sources to optimize API costs and data freshness.

Kaggle & CSV Datasets

Bulk ingestion, auto-mapped columns, high volume, static historical baseline. Highly cost-effective.



Community User Reviews

Direct user submissions via the platform. High trust (Verified Buyer status), builds a proprietary localized database over time.



Live Web & Synthetic AI

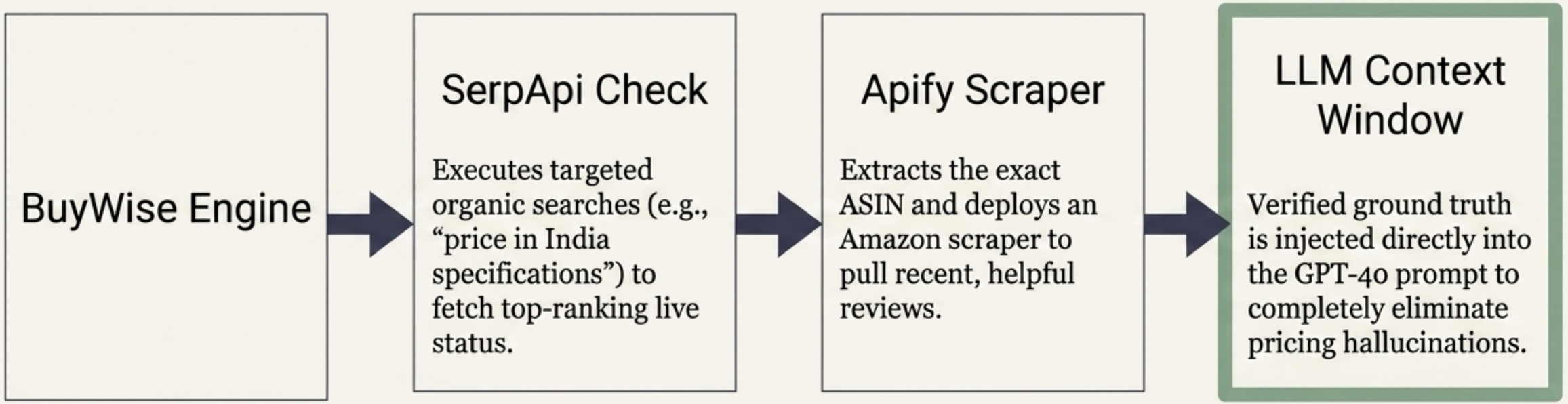
Triggered only when local data is sparse. Synthesizes contextual reviews via GPT-4o-mini and grounds pricing in real-time internet data. Ensures zero dead ends for searches.



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Real-Time Market Grounding

The Problem: LLMs suffer from outdated training data, routinely hallucinating current Indian Rupee (₹) pricing and failing to realize when tech is discontinued.

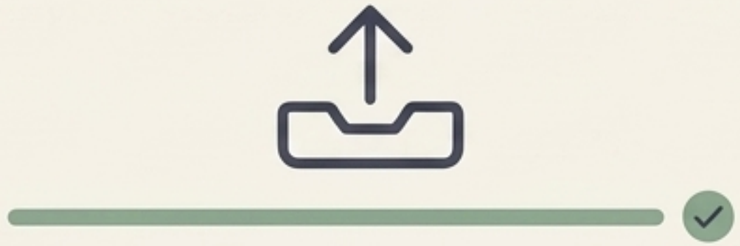


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Intelligent Admin Data Hub

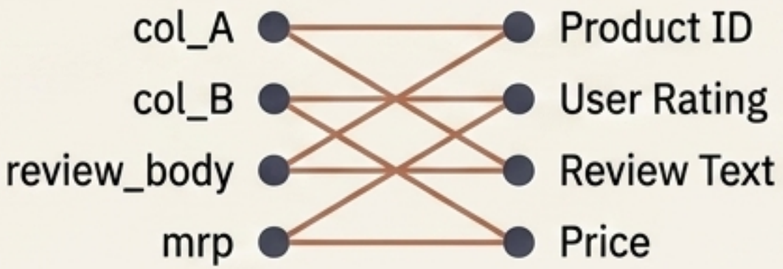
1. Smart Ingestion

A seamless 2-step CSV/Excel upload flow built for massive historical datasets.



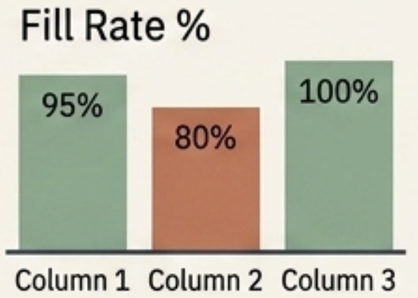
2. Auto-Mapping Intelligence

Automatically detects and assigns semantic roles to raw columns, accurately recognizing ‘review_body’ as Review Text, or ‘mrp’ as Price.



3. Automated Data Profiling

Before confirming ingestion, the system infers column types, calculates fill rates (%), counts unique values, and provides a 5-row sample preview to ensure strict database integrity.



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System Architecture & Design Stack

Frontend (The Glass Interface)

HTML5/CSS3 custom variables, GSAP ScrollTrigger for complex scroll animations, Three.js Three.js for 3D elements. Typography via Space Grotesk and Inter.

Backend (The Engine)

Python 3.9+, Flask, and SQLite with Werkzeug password hashing for secure authentication.

AI & Data (The Brain)

OpenAI API (gpt-4o / gpt-4o-mini), scikit-learn for TF-IDF and Cosine Similarity, NLTK for VADER Sentiment Analysis, Apify, and SerpApi.

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Phase 2 & The Ecosystem Horizon

Deep E-Commerce Integration

Full product detail pages with granular review breakdowns and side-by-side spec comparison matrices.

Proactive Intelligence

Implementing active price tracking and drop alerts specifically calibrated for Indian marketplaces.

Seamless Identity

Google and GitHub OAuth integration for frictionless community onboarding.

The BuyWise Extension

A browser extension that overlays Trust Scores and Fake Review warnings directly on top of Flipkart and Amazon pages while the user shops.

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